ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO ROBOTICS

PAPER 1 · PRACTICE E · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Introduction to Robotics paper 1 · practice e.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and wording that shows understanding.

SECTION A

QUESTION 1

1 1.1 A ✓ 1.2 C ✓ 1.3 B ✓ 1.4 D ✓ 1.5 B ✓ 1.6 A ✓ 1.7 D ✓ 1.8 C ✓ 1.9 A ✓ 1.10 B ✓ (10)

[10]

QUESTION 2

2 2.1 True ✓ 2.2 False ✓ 2.3 True ✓ 2.4 False ✓ 2.5 True ✓ (5)

[5]

QUESTION 3

3 3.1 D ✓ 3.2 E ✓ 3.3 I ✓ 3.4 G ✓ 3.5 B ✓ 3.6 H ✓ 3.7 A ✓ 3.8 J ✓ 3.9 C ✓ 3.10 F ✓ (10)

[10]

QUESTION 4

4 4.1 Printed Circuit Board ✓ 4.2 Computer-Aided Design ✓ 4.3 Revolutions Per Minute ✓ 4.4 Passive Infrared ✓ 4.5 Liquid Crystal Display ✓ (5)

[5]

TOTAL SECTION A: [30]

SECTION B

QUESTION 5

5.1 It senses changes in the infrared (heat) radiation from warm bodies ✓ when a person or animal moves across its field of view ✓; its output goes HIGH when movement is detected ✓. Use ✓: a security / patrol robot detecting intruders, a robot that greets people, waking a robot from sleep. (4)

5.2 A sensor-navigating robot measures its surroundings (distance, lines, obstacles) and changes its path as needed ✓✓. A pre-programmed robot follows set moves (e.g. drive 2 m, turn 90°) without checking its surroundings, so it cannot react to obstacles or errors ✓✓. (4)

5.3 The battery voltage is higher than the electronics need and drops as the battery runs down ✓✓; the regulator gives a steady voltage (e.g. 5 V) so the microcontroller and sensors work correctly and are not damaged ✓✓. (4)

5.4 CAD (Computer-Aided Design) is software used to design parts on a computer ✓; the part is drawn in 3D with exact dimensions and checked for fit ✓; it is then exported (STL) ✓ and sliced for the 3D printer ✓. (4)

5.5 Any FOUR ✓✓✓✓: they do dangerous jobs (welding, painting, handling chemicals); they lift heavy loads; they work in hot, toxic or radioactive places; they inspect mines, pipes and tunnels; they reduce repetitive-strain injuries; bomb disposal. (4)

[20]

QUESTION 6

6.1 A gear reduction (e.g. 48 : 1) makes the wheel turn more slowly than the motor ✓✓ but with more torque (turning force) ✓✓, so the robot can carry weight and climb slopes. (4)

6.2.1 $C = \pi \times d = \pi \times 6{,}5 \checkmark = 20{,}42 \text{ cm} \checkmark$ (2)

6.2.2 $v = (120 \div 60) \times 20{,}42 \checkmark = 40{,}84 \text{ cm/s} \checkmark$ (2)

6.3 $I = 3 \times 20 = 60 \text{ mA} \checkmark (0{,}06 \text{ A}) \checkmark$ (2)

6.4 $t = 1{,}5 \text{ Ah} \div 0{,}5 \text{ A} \checkmark = 3 \text{ h} \checkmark$ (2)

6.5 $P = V \times I = 6 \times 0{,}75 \checkmark = 4{,}5 \text{ W} \checkmark$ (2)

6.6 How much charge the battery stores ✓ – e.g. 1 500 mAh can supply 1 500 mA for one hour (or 750 mA for two hours) ✓. (2)

[16]

QUESTION 7

7.1 So the drone knows its exact position ✓ and can fly the planned route over the field by itself ✓. (2)

7.2 It detects which plants are unhealthy ✓ so the pump sprays only those areas instead of the whole field ✓. (2)

7.3 The propellers and the heavy spray tank use a lot of power ✓, so flight time is short and the drone must land often to recharge or swap batteries ✓. (2)

7.4 Any TWO ✓✓: register the drone / get a licence (SACAA rules); keep it away from people, houses and roads; fly only in good weather; keep it within sight; handle the chemicals safely (PPE). (2)

7.5 Any ONE ✓✓: faster; less chemical used; workers are not exposed to the spray; reaches areas that are hard to walk through; works without damaging the crops. (2)

[10]

QUESTION 8

8.1 It sets all four motor-driver inputs in one call ✓✓, so each movement (forward, reverse, turn) is one short line instead of four ✓, which makes the code shorter and easier to read and change ✓. (4)

8.2 It adds 1 to the bump counter ✓; it reverses both wheels for 0,4 s ✓✓; then it turns right (left wheel forward, right wheel back) for 0,3 s, away from the obstacle ✓. (4)

8.3 INPUT_PULLUP switches on the internal pull-up resistor ✓, so the input is HIGH when the bumper is not pressed and never "floats" ✓; the switch connects the pin to GND, so digitalRead() gives LOW when it is pressed ✓✓. (4)

8.4 $\text{Duty cycle} = 180 \div 255 \times 100$ ✓✓ $= 70{,}59\%$ ✓✓ (4)

8.5 Both motors stop ✓, the suction fan is switched off ✓, the message "Bin full - please empty" is sent to the serial monitor ✓, and the endless while loop stops the program until the robot is reset ✓. (4)

8.6 Any sensible improvement with an explanation ✓✓✓✓, e.g. add an ultrasonic / IR sensor to slow down before hitting obstacles; use a random turn angle so the robot covers the room better; add a battery-level check; add a bin-full sensor instead of counting bumps; use millis() instead of delay() so the bumpers are still checked while turning. (4)

[24]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE